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20 April 1950
Superseding
AN-C-96
10 February 1944

MILITARY SPECIFICATION

COLLIMATOR; SEXTANT, AIRCRAFT

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers the requirements for one type of sextant collimator.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein.

2.1.1 Specifications.-

Federal

QQ-P-416 Plating; Cadmium (Electrodeposited)

Military

MIL-P-6064 Packaging of Lightweight Aircraft Accessories

Air Force-Navy Aeronautical

AN-QQ-A-696 Anodic Films; Corrosion-Protective (for) Aluminum Alloys
AN-P-32 Plating, Zinc
AN-P-39 Plating; Chromium
AN-S-126 Screw-Threads; Standard, Aircraft

U. S. Army

94-40645 Marking; Exterior, Domestic and Export Shipment, by Contractors 1/

2.1.2 Other Publications.-

Bureau of Supplies and Accounts

Navy Shipment Marking Handbook 2/

2.1.3 Drawings.-

Air Force-Navy Aeronautical Standard Drawing

AN5853 Collimator - Sextant

1/ Applicable only to Air Force purchases.
2/ Applicable only to Navy purchases.

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(Copies of this specification and copies of other publications referenced herein or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio; or the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials.- Materials shall conform to applicable specifications as specified herein. Materials which are not specifically designated shall be of the best commercial quality and suitable for the purpose intended.

3.1.1 Metals.- Metals shall be suitably protected to resist corrosion during normal service life.

3.2 Design and Construction.- The collimator shall conform to Drawing AN5853. Construction shall be such as to withstand normal strains of jars, vibrations, and such other conditions as are incident to shipping, storage, and service, without failure.

3.2.1 Base.- The base shall be made of cast aluminum-alloy ribbed underneath to obtain lightness and rigidity.

3.2.1.1 Leveling Screw.- When the leveling screw is in its mean position the bottom of the "V'S" of the base casting shall be approximately parallel to the surface on which it is standing.

3.2.2 Lens.- The collimator lens shall be made of the best grade optical glass, shall be free of spherical and chromatic aberration and other distortion, and shall be accurately ground and polished as required to ensure conformance with the requirements of this specification. The effective aperture of the collimator objective shall be not less than 42 millimeters.

3.2.3 Reticle.- It shall be possible to illuminate the reticle by means of an external light source not a part of the instrument.

3.2.3.1 Adjustments.- The reticle and the objective lens of the collimator shall be capable of adjustment to enable correction of any possible shift of alinement or focus.

3.2.4 Levels.-

3.2.4.1 Straddle Level.- A straddle level shall be furnished for leveling the collimator. The level shall be designed to rest on the collimator by means of a V-slot in each end which shall rest on the tube lands. It shall be possible to reverse the level.

3.2.4.2 Cross Level.- A cross level shall be mounted on the straddle level.

3.3 Interchangeability.- Parts of all collimators of a given manufacturer's part number shall be interchangeable with parts of collimators carrying the same manufacturer's part number. The manufacturer's part number is intended for reference in the purchase of spare parts for the collimators. If it is necessary to make any change which would affect the interchangeability of parts, the manufacturer's part number of the part and of the collimator shall be changed.

3.4 Protective Coatings and Finishes.- Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes in atmospheric conditions shall not be used.

3.4.1 Aluminum-Alloy Parts.- Aluminum alloy parts shall be covered with an anodic film conforming to Specification AN-QQ-A-696, except that small holes need not be anodized.

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3.4.2 Steel Parts.- Steel parts shall be cadmium-, zinc-, or chromium-plated, where practicable, in accordance with Specification QQ-P-416, and AN-P-32, or AN-P-39, respectively.

3.4.3 Exposed Metallic Parts.- Except as otherwise specified, exposed metallic parts shall be coated with durable black baked enamel.

3.5 Screw Threads.- Unless otherwise specified, screw threads shall be in accordance with Specification AN-S-126.

3.6 Accessories.-

3.6.1 Carrying Case.- The carrying case shall be suitably constructed to insure adequate protection to the collimator against jars, shocks, vibrations, and other conditions incident to handling, shipping, storage, and service use. Special care shall be taken that the collimator tube lands and the V-slots in the base are completely protected in order to prevent misalinement of these parts. The case shall be made of a suitable wood finished on the inside with two coats of varnish and on the outside with one coat of grey enamel, and shall be equipped with metal handle, hinges, and lock, suitably protected against corrosion. All surfaces of contact between the collimator and the inside of the case shall be fully covered with a suitable wool felt 1/8-inch thick. A rubber bumper shall be located at each corner on the outside bottom of the case.

3.6.2 Tools.- Tools necessary for the adjustment of the collimator shall be supplied with each collimator and securely mounted in the carrying case.

3.6.3 Instruction Pamphlet - A pamphlet, covering the operation and adjustments of the collimator, shall be furnished with each collimator.

3.6.3.1 Envelope.- The pamphlet shall be contained in an envelope marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS"

3.7 Identification of Product.-

3.7.1 Nameplate.- A nameplate shall be securely attached to the exterior of the carrying case and shall be legibly marked with the following information:

COLLIMATOR - SEXTANT
AN5853- (proper dash No.)
Manufacturer's Part No.
Order No. (or Contract No. if Order No. is not assigned)
Manufacturer's name or trade-mark (omit address)

3.8 Performance.- The collimator shall satisfy the performance requirements specified in Section 4 under paragraphs headed as follows:

(a) Centering
(b) Parallelism
(c) Straddle Level Sensitivity
(d) Cross Level Sensitivity

3.9 Workmanship.- All details of workmanship shall be in accordance with high grade optical instrument manufacturing practice.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- All the tests required for the testing of collimators are classified as Inspection tests, for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Inspector. The contractor shall maintain a record, available to the Inspector, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.2 Individual Tests.- Each collimator submitted for acceptance under contract shall be subjected to the following tests. In addition, each collimator shall be subjected to any other test specified herein which the Inspector considers necessary to determine conformance with the requirements of this specification.

4.2.1 Examination of Product.- Each collimator shall be carefully examined to determine conformance with this specification with respect to requirements not covered by tests.

4.2.2 Centering.- The collimator tube shall be mounted in the V-slots of the base and rotated about its longitudinal axis, and the image of the collimator reticle observed with a suitable optical device. The image of crossline intersection of the reticle shall not move from its mean position more than a distance equivalent to the displacement obtained by tilting the optical device through 1 minute of arc.

4.3 Sampling Tests.- At least 3 from every 50 collimators and samples selected at random and as often thereafter as the Inspector deems advisable shall be subjected to any or all of the following tests.

4.3.1 Parallelism.- A mask having a 1/8-inch aperture shall be moved across the collimator lens, and the image displacement shall be determined by the use of a suitable optical device. The image of crossline intersection of the reticle shall not move from its mean position more than a distance equivalent to the displacement obtained by tilting the optical device through 30 seconds of arc.

4.3.2 Straddle Level Sensitivity.- The straddle level shall be set upon a tilting base with a suitable reference level of known sensitivity. The tilting base shall be moved through an angle necessary to produce a 2-millimeter movement of bubble. The angle shall be between 20 and 40 seconds of arc.

4.3.3 Cross Level Sensitivity - The test specified for "Straddle Level Sensitivity" shall be conducted on the cross level. The angle shall be between 5 and 10 minutes of arc.

4.4 Acceptance Stamp.- Each accepted collimator shall be marked with the Inspector's acceptance stamp.

4.5 Rejection and Retest.- When any representative sample fails to meet the requirements of the Sampling tests, the lot represented shall be rejected and returned at the contractor's expense. Any collimator failing to meet the requirements of the individual tests shall be rejected and returned at the contractor's expense. Collimators which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action take to correct the original defects shall be furnished the Inspector. Collimators rejected after retest shall not be resubmitted without the specific approval of the Procuring Service.

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5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation, Packaging, and Packing.- Preservation, packaging, and packing shall be in accordance with Specification MIL-P-6064, Group VI.

5.3 Marking and Labeling.-

5.3.1 Packages.- Each interior package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened

COLLIMATOR - SEXTANT
AN5853- (proper dash No.)
Manufacturer's Part No.
Name of Contractor (and name of Manufacturer if not the same)
Government Order No. (or Contract No. if Order No. is not assigned)

5.3.1.1 Acceptance Stamp.- Each package shall be marked with the Inspector's acceptance stamp, imprinted partly on the sealing strip and partly on the carton.

5.3.2 Shipping Containers.- Each shipping container shall be marked in accordance with the requirements applicable to the individual Services as specified in Specification 94-40645 for the Air Force, and in the Navy Shipment Marking Handbook for the Navy.

5.4 Precautionary Markings.- The following precautionary marking shall appear on two opposite sides of each interior package and on the shipping containers:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended Use.- Sextant collimators covered by this specification are intended for use in checking sextant index errors, and are interchangeable with those covered by Navy Aeronautical Specification C-102 and ANA Specification AN-C-96.

6.2 Ordering Data.- Requisitions, contracts, and orders should state the AN part number of the collimator desired.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Navy - BuAer

Other interest:
Air Force

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